A Brain-Inspired Vision for Planetary-Scale Cooperative 7G Backhaul

Active-Active Multi-Medium Content Transport, Self-Scaling Content, and Orbital AI Edge Infrastructure

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Abstract

Future communication systems may need to coordinate terrestrial, aerial, and space-based networks as one service fabric, even when individual links vary greatly in capacity, delay, cost, availability, and packet size. Building on the author's master's thesis *Gezora: A Self-Scalable Content Distribution Network* (2010), this paper presents a brain-inspired allegory for AI-coordinated backhaul: network telemetry acts as distributed sensing, caches provide distributed memory, forecasting anticipates changing conditions, and routing and scaling policies act on the environment to maintain service. The analogy organizes a long-horizon vision for cooperative, active-active backhaul in which multiple heterogeneous paths can serve an application concurrently and content replicas move toward predicted demand. A synthetic public-safety study makes real-time voice/video on-time delivery, jitter, freeze duration, and interruption during medium changes primary service outcomes rather than relying on throughput alone. Orbital compute is considered as a possible edge tier for selected workloads, not as a presumed replacement for terrestrial data centers. The design is a distributed engineering architecture—not a biological brain, sentient system, or demonstrated autonomous global network. It treats quantum networking as a separate resource plane rather than a byte-carrying IP path, and distinguishes application-session continuity from an unattainable guarantee of zero packet delay or loss. A second synthetic experiment compares reactive multipath with forecast-assisted path activation and multicloud replica preparation under link and cloud failures; its results are conditional on explicitly modeled controller delays and resource assumptions.

The paper argues for a mathematical, time-varying, multilayer topology model to coordinate services across heterogeneous networks. This model is proposed as a cross-layer coordination and assurance plane alongside OSI and existing protocol stacks—not as a replacement for their standardized interfaces or encapsulation. Its hypothesized benefits in predictability, security, and

interoperability require formal specification and comparative validation. The paper connects this thesis-inspired idea to multi-path transport, delay-tolerant networking, edge inference, and space sustainability. It presents a transparent control objective and an evaluation agenda, including a synthetic public-safety handover study of acceleration-triggered cross-medium overflow, rather than claiming demonstrated planetary-scale performance. The accompanying repository is an educational prototype: its cooperative failure benchmark uses deterministic forecast logic, not trained distributed AI or live Kubernetes. It models mesh allocation and cache decisions, verifies UDP transmission on local loopback, and provides an ONNX congestion forecaster pretrained on synthetic data. It does not yet implement the proposed topology model, transmit over a real multi-medium network, or execute on an NPU in space or on Earth.

Keywords: heterogeneous backhaul; multipath networking; content delivery; congestion forecasting; edge computing; orbital computing; delay-tolerant networking; network sustainability

1. Introduction

Network users experience a service session, while the infrastructure beneath that session changes continually. A device may move between Wi-Fi and cellular coverage; a rural or maritime service may depend on microwave, satellite, or radio links; and an application may retrieve content from an edge cache, a cloud, or a more distant origin. These are related but distinct control problems: preserving identity and application state, choosing paths, adapting to link constraints, and deciding where content or computation should reside.

The conceptual root is Nikolic's master's thesis *Gezora: A Self-Scalable Content Distribution Network* (2010) [1]. In this paper, the author frames that self-scaling CDN research as an early attempt to model an **artificial-brain-like network**: a system that observes requests and network conditions, retains useful content, anticipates future demand, and changes where service resources are placed. This is an allegory for coordinated feedback and adaptation. It does not imply that the thesis implemented modern AI, that the network is conscious, or that a biological brain is an engineering blueprint.

This paper advances a vision of **cooperative 7G backhaul**: a planetary-scale service fabric in which terrestrial, aerial, and orbital nodes cooperate to deliver data and compute using available resources while adapting to changing demand and network conditions. Here, "7G" names a research vision beyond standardized 6G work; it is not an approved generation, specification, or forecast of a particular deployment date. ITU-R Recommendation M.2160 defines the framework and objectives for IMT-2030, providing a standards-oriented context for future system research [2].

The motivating principle comes from self-scalable content distribution: observe demand, learn where it is growing, and move capacity and content closer to users [1]. The extension proposed here is to make the transport fabric part of that adaptation. If congestion is rising rapidly on one route, traffic and replication decisions should react before the link is unusable, while remaining bounded by cost, energy, cache capacity, policy, and reliability constraints.

The vision's ambitious goal is broader access to useful communication and computation, including underserved and remote regions. This is a research objective, not a claim that the proposed design will necessarily improve global communications. Achieving that outcome would require technical validation, interoperability, affordability, security, regulatory coordination, and demonstrated environmental responsibility.

2. Design principles

The architecture is guided by six principles:

1. **Service continuity, not "magic handover."** Preserve application identity, session state, and reassembly across access changes. Do not equate this with zero interruption, zero loss, or uninterrupted radio coverage.
2. **Active-active when beneficial.** Use more than one eligible path concurrently when it improves service, but permit one path when parallel use increases cost, energy, reordering, or risk without useful benefit.
3. **Forecast before reacting.** Consider congestion level and its temporal velocity and acceleration, with uncertainty and conservative safety bounds.
4. **Scale content and computation together.** Place cache replicas and eligible workloads near predicted demand, subject to origin, capacity, residency, privacy, and egress constraints.
5. **Keep control explainable and bounded.** Separate policy recommendations from execution; verify proposed actions against hard constraints and retain operator override and auditability.
6. **Use each medium according to its properties.** Radio, fiber, satellite, optical links, and quantum resources have different failure modes and semantics; they are not interchangeable byte pipes.

2.1 The artificial-brain allegory

The brain analogy provides a high-level way to explain how the proposed network coordinates sensing, memory, prediction, and action. It is an architectural metaphor for distributed control, not a claim of biological equivalence or general intelligence.

Allegorical function	Network counterpart	Intended engineering role
Sensation	Link telemetry, request arrivals, cache outcomes, and node health	Observe local and shared operating conditions
Memory	Verified edge caches, bounded telemetry histories, and model state	Retain useful content and information needed for adaptation
Anticipation	Demand and congestion forecasts, including velocity and acceleration	Estimate near-future conditions and uncertainty

Allegorical function	Network counterpart	Intended engineering role
Coordination	Peer policies and federated control exchanging authorized summaries	Coordinate traffic and placement without requiring one global brain
Action	Path-share changes, cache replication, workload placement, and scaling requests	Apply bounded, auditable resource changes
Homeostasis	Availability constraints, hysteresis, redundancy, repair, and safe fallback	Maintain service objectives as individual components vary or fail
Learning	Evaluation of decisions against measured outcomes	Improve policies only within explicit safety and governance limits

This mapping makes the thesis-inspired idea concrete: self-scaling is not only adding resources when demand crosses a threshold; it is a feedback process that couples demand memory, prediction, transport decisions, and placement. "Self-healing" means detecting a fault and restoring an explicitly defined service objective when alternate resources exist. It cannot restore a severed network where no route or stored content remains. "Self-scaling" is bounded by physical capacity, power, cost, operator policy, and environmental constraints.

The biological analogy is inspiration, not evidence that a network will work. Acquired savant abilities after central-nervous-system injury have been described in rare clinical cases and reviewed in the literature [15]. Those observations do not establish that injury generally causes beneficial rewiring, that savant abilities are a predictable result of brain reorganization, or that biological mechanisms transfer directly to routers. The engineering analogue is narrower and testable: after an injected path, router, edge-site, or cloud-region failure, measure whether the system detects the damage, reallocates traffic and eligible workloads over surviving resources, and restores declared QoS without violating security, cost, or stability constraints. No injury-to-savant mechanism is assumed in the model.

The allegory also emphasizes decentralization. A planetary system cannot assume continuous control-plane connectivity, so local nodes should sense and act safely using local state. Coordination among nodes may improve global efficiency, but must tolerate delayed, missing, or conflicting information. This leads to testable properties—response time, stability, continuity, resource efficiency, and safe fallback—rather than anthropomorphic claims.

2.2 Comparison with 6G mobile plus backhaul

"6G mobile network plus backhaul" is not one fixed topology or a finished product architecture. ITU-R defines the IMT-2030 framework and objectives; 3GPP Release 20 contains 5G-Advanced work and early 6G studies, while Release 21 is identified by 3GPP as the first normative 6G release [2,

14]. As of this paper's date, a fair comparison is therefore between a **6G-oriented mobile system using operator-managed backhaul** and the **proposed cooperative backhaul overlay**, not between deployed 6G and an established 7G standard. Existing networks may already use redundant paths, multi-access steering, edge compute, automation, and AI/ML. The proposal does not claim these individual ingredients are new or absent from 6G research.

Dimension	6G mobile system plus managed backhaul (research/reference architecture)	Proposed 7G cooperative multipath backhaul
Scope	6G radio/core and associated transport/backhaul, built from standardized functions and operator policies as they mature	Cross-domain coordination overlay spanning access, terrestrial/aerial/orbital backhaul, caches, edge sites, and multiple clouds
Path use	May use redundancy, multi-access steering, or multipath where protocols, endpoints, and operator configuration support it; not assumed to be single-path	Treat eligible heterogeneous paths as a shared active-active resource pool; stripe or duplicate deadline-sensitive traffic when policy and capacity permit
Congestion response	Operator control and automation can react or predict within deployed system capabilities; there is no single universal 6G backhaul algorithm	Forecast congestion level, velocity, and acceleration; pre-activate and rebalance routes before a threshold, with bounded hysteresis and cost/energy budgets
Self-reorganization	Resilience and automated management are valid design goals; specific recovery behavior depends on implementation	Brain-inspired functional loop: detect damage, assess surviving topology, re-route traffic, move/replicate eligible content or workloads, then verify service objectives
AI and control	AI/ML is a 5G-Advanced and 6G research/work area, not proof that all networks use autonomous AI routing	Cooperative AI may recommend cross-layer actions, but deterministic admission, safety limits, audit, and operator override constrain execution

Dimension	6G mobile system plus managed backhaul (research/reference architecture)	Proposed 7G cooperative multipath backhaul
Media QoS	Can provide service-specific QoS depending on supported transport and deployment; no universal guarantee across every access combination	Explicitly optimize application deadlines, freezes, jitter, and interruption, with selective duplication/FEC overhead measured
Cloud and edge	Mobile edge and NTN integration are within the broader standards and deployment landscape	Coordinate path choice with cache placement, multicloud workload placement, and bounded Kubernetes scale-out; these are slower control loops than packet routing
Evidence status	Future 6G requirements and specifications are in development; implementation will vary	Research vision and partial simulator prototype; not a standard, deployed network, or validated production design

The proposed distinction is consequently **system-level cooperation and measurable policy**, not the mere presence of AI, multipath, satellites, or Kubernetes. A 6G operator could implement some of the proposal's functions; that would be evidence of overlap and interoperability potential, not proof that the complete architecture is already standardized or universally available. Conversely, labeling a system "7G" does not establish novelty: novelty must be demonstrated by a precise mechanism, baseline, and repeatable QoS/resource results.

2.3 Closest related research

Several peer-reviewed studies directly overlap with the transport and media parts of this proposal. Wu et al. evaluate high-quality mobile video over heterogeneous MPTCP paths, including video-quality and FEC decisions [19]. Corbillon et al. use video-layer information to schedule data over MPTCP [20], and Lim et al. evaluate an earliest-completion-time MPTCP scheduler for heterogeneous paths [21]. Qin et al. specifically study reinforcement-learning scheduling for heterogeneous MPTCP [22]. Kreith and Drozdy evaluate rate-controlled real-time media delivery over multiple paths [23]. These are strong precedents for heterogeneous multipath, video-aware scheduling, learning-based path scheduling, and media QoS. DetNet's packet replication and elimination architecture is a further standards-level precedent for redundant real-time delivery [24].

The network/compute coupling also has direct precedents. Poularakis et al. jointly optimize edge-service placement and request routing across multi-cell mobile-edge networks [25]. FluidRAN jointly considers virtualized RAN-function placement and routing paths while accounting for MEC

services [26]. These studies overlap with the proposed joint placement/routing objective, but do not evaluate simultaneous heterogeneous multipath, media packet duplication, or the proposed forecast-triggered recovery loop.

The closest studies address separate pieces of the vision; their reported results do not validate this repository's forecast thresholds, congestion velocity/acceleration signal, selective packet duplication, or combined network/content/cloud control loop. In this focused comparison, no cited study evaluates that full combination. This is not a claim that no such work exists or a proof of novelty: a systematic search across networking, multimedia, edge/cloud orchestration, and 3GPP/ETSI work is still required. General AI-for-networking surveys and generic edge-orchestration references are intentionally omitted here because they are not close enough to count as direct precedents.

3. Beyond layer-centric coordination: a mathematical topology model

OSI and Internet protocol layering provide essential abstractions for standardizing communication functions and interoperable protocol interfaces [8, 9]. They should not be discarded. The research concern is narrower: a stack of protocol layers and encapsulated packets does not, by itself, express the changing end-to-end relationships among radio links, routes, caches, compute services, trust domains, orbital contacts, and application objectives. In a self-scaling system, decisions about one of these resources may affect several others at once. Cross-layer signaling and policy can address parts of this problem, but a common, explicit model of those relationships is still needed for the vision proposed here.

The paper therefore proposes a **temporal, typed, multilayer service topology** as a coordination model. This is a research hypothesis, not a result showing that OSI is inadequate or that a mathematical graph automatically improves a network. The graph describes which services and resources can interact, under what conditions and trust relationships; conventional protocols continue to carry the actual bits and preserve their own encapsulation and wire formats. Multilayer network theory offers established vocabulary for representing systems with multiple kinds of nodes and relations [10].

At time t, define:

$$\mathcal{G}(t) = (V(t), E(t), \Lambda, \tau(t), \eta(t), \rho(t), \Pi(t)),$$

where $V(t)$ contains typed entities such as devices, links, caches, compute sites, and services; $E(t)$ contains directed relations such as connectivity, forwarding, service dependency, replication, and control; Λ labels the relation layers; $\tau(t)$ records temporal state and contact availability; $\eta(t)$ carries measured or estimated properties such as capacity, delay, loss, cost, energy, MTU, queue state, and uncertainty; $\rho(t)$ records identity and trust claims with provenance; and $\Pi(t)$ contains policy and service constraints. Higher-order relations may be needed when a decision depends jointly on several

resources, such as a service that requires a compute site, a scheduled satellite contact, and a ground station.

Unlike a static network map, $\mathcal{G}(t)$ changes as links appear or fail, contacts open or close, queues evolve, and caches or compute capacity scale. Each observation and forecast should carry a timestamp, source, validity interval, and uncertainty. This allows the controller to distinguish a measured link from a predicted state and to reject stale or untrusted information.

An end-to-end service path P is admissible only if all required predicates hold:

$$\begin{aligned}\text{Admissible}(P,t) = \ & \text{Reachable}(P,t) \wedge \text{CapabilitiesMatch}(P) \\ & \wedge \text{PolicyAllows}(P,t) \wedge \text{TrustValid}(P,t) \\ & \wedge \text{ResourceFeasible}(P,t).\end{aligned}$$

The predicates cover reachability, compatible service capabilities, data residency and authorization, verified identity and telemetry, MTU and capacity, and any application deadline or availability requirement. A controller could then optimize a declared objective, for example:

$$\min_{x \in \mathcal{F}(t)} \left[w_d D(x,t) + w_c C(x,t) + w_e E(x,t) + w_r R(x,t) \right],$$

where x is a set of path, cache, and compute allocations; $\mathcal{F}(t)$ is the set of allocations satisfying the admissibility predicates and hard capacity and policy constraints; and D, C, E, R denote delay, monetary cost, energy, and risk. Weights w_* express operator or application policy and must not override hard safety, privacy, or regulatory constraints. The objective is a starting point for formalization, not a solved optimization algorithm.

3.1 Intended properties and limits

- **Predictability:** temporal reachability, path diversity, cut sets, scheduled contact windows, and resource bounds can be analyzed over the topology. Performance prediction additionally requires traffic, queueing, channel, and uncertainty models; topology alone cannot guarantee latency or delivery.
- **Security:** trust and policy labels can make authorization conditions explicit and machine-checkable. They do not create authentication, confidentiality, or integrity. Those require cryptographic protocols, credential lifecycle, authenticated telemetry, implementation assurance, and a defined threat model.
- **Interoperability:** a common semantic model can describe service capabilities and constraints across adapters. Actual interoperability still depends on agreed schemas, protocol bindings, conformance tests, governance, and backward-compatible mappings to deployed standards. A graph schema alone cannot make independently implemented systems interoperate.
- **Encapsulation:** each adapter maps between the service-topology model and existing protocol operations. Packet headers, MTU, tunnel overhead, security encapsulation, and transport

semantics remain properties of the underlying protocols and paths; they must not be hidden by an abstract edge.

The distinction is important: the proposed topology model is a coordination, prediction, and assurance layer **over** standardized protocol stacks, not a new universal packet format intended to replace Ethernet, IP, transport, or application protocols. This separation is intended to preserve proven interoperability while giving the self-scaling controller a global view of dependencies and constraints. Whether the model delivers better outcomes is an open research question.

3.2 Evaluation hypotheses

The following are falsifiable hypotheses, not established conclusions:

1. Relative to independently optimized protocol layers, a typed temporal topology controller will reduce service-SLO violations under multi-medium failures without increasing control instability.
2. Explicit capability, policy, trust, and provenance predicates will reduce invalid placement and unauthorized-path decisions compared with implicit cross-layer state, while keeping adapter overhead measurable.
3. Standardized adapters and conformance suites will achieve interoperability across heterogeneous implementations without requiring identical internal topology representations.

Evaluate these hypotheses against a layer-local baseline and a simpler cross-layer policy. Report SLO violation probability, reachability and continuity, prediction error, failover time, path reordering, cache hit/cost, energy per delivered byte, policy violations, control-plane overhead, and cross-vendor conformance results. Inject stale telemetry, forged measurements, schema mismatch, partial partitions, and unavailable resources. The model should fail closed for authorization, degrade predictably for missing measurements, and retain deterministic fallback. Topological invariants can support proofs about connectivity or policy reachability only when the model and assumptions correspond to the deployed system.

4. Proposed planetary architecture

The design is a federation of cooperating nodes rather than one global controller:

- **Access and edge nodes** serve mobile, fixed, rural, maritime, and infrastructure users. They may host local caches, session anchors, and low-latency inference.
- **Terrestrial backhaul nodes** connect fiber, microwave, cellular, Wi-Fi, HF, and other operator-provided links. Link adapters expose common telemetry while preserving medium-specific limits.
- **Aerial relays** may provide temporary or mobile coverage. Their energy, capacity, position, and availability are explicit resource constraints.
- **Orbital relay and compute nodes** may provide inter-satellite transport, store-carry-forward service, or selected edge workloads. They participate only when their contact schedule, capacity,

thermal state, power budget, and round-trip delay fit the application.

- **Cloud and AI data centers** provide large-scale storage, training, and compute. Workload placement considers network and compute conditions rather than assuming that the nearest network node has sufficient accelerator capacity.
- **A policy and trust plane** defines authorization, identity, data locality, model provenance, and safety bounds. Local nodes can continue operating during control-plane partition, subject to conservative policy.

The control plane observes topology and telemetry, forecasts demand and link conditions, proposes route and cache actions, and checks them against policy. The data plane forwards packets or bundles using installed, validated actions. Fast local protection must not depend on a remote AI inference service. Models may recommend slower-timescale changes, while deterministic fallback policies handle link loss, stale models, missing telemetry, and partitions.

For long or intermittent paths, an IP session alone may not be the right delivery abstraction. Delay-Tolerant Networking's Bundle Protocol v7 is an existing standards-track store-carry-forward approach for stressed environments with intermittent connectivity, long or variable delay, and high error rates [3]. A future architecture could combine IP multipath for connected operation with DTN bundles for delay-tolerant delivery; the two modes must not be misrepresented as identical service.

5. Congestion-aware active-active allocation

Let link i have utilization $u_i(t)$, queue delay $q_i(t)$, loss estimate $\ell_i(t)$, capacity c_i, and base latency d_i. A normalized pressure indicator can combine these measurements:

$$p_i(t) = \operatorname{clip}\left(0.70u_i(t) + 0.20\frac{q_i(t)}{d_i + q_i(t) + 1} + 0.10\ell_i(t),\, 0,\, 1\right).$$

For sample interval Δt, estimate pressure velocity and acceleration:

$$v_i(t) = \frac{p_i(t) - p_i(t - \Delta t)}{\Delta t}, \qquad a_i(t) = \frac{p_i(t) - 2p_i(t - \Delta t) + p_i(t - 2\Delta t)}{(\Delta t)^2}.$$

A transparent baseline forecasts pressure over horizon H:

$$\hat{p}_i(t + H) = \operatorname{clip}\left(p_i(t) + v_i(t)H + \frac{1}{2}a_i(t)H^2,\, 0,\, 1\right).$$

This polynomial is a baseline predictor, not a claim that congestion follows a constant-acceleration law. A deployable policy should estimate uncertainty, handle irregular samples, detect regime shifts, and compare the baseline with learned predictors.

5.1 Congestion waves and threshold overflow

The proposed "congestion wave" analogy describes a rising disturbance in offered load, queue occupancy, or normalized path pressure. It is not a claim that network congestion follows seismic physics. In particular, delivered throughput can flatten at a bottleneck, so a controller should forecast offered demand and pressure telemetry rather than infer an approaching limit from delivered throughput alone.

For normalized utilization $u_i(t)$, estimate its velocity and acceleration from samples and forecast utilization over a control horizon:

$$\hat{u}_i(t+H) = \mathrm{clip}\left(u_i(t) + H\dot{u}_i(t) + \frac{1}{2}H^2\ddot{u}_i(t), 0, 1\right).$$

When forecast utilization approaches a configured safety threshold, the controller can reserve or activate a secondary medium before the current path fills. If forecast aggregate demand exceeds the active paths' safe capacity, traffic is progressively spilled across additional eligible media, subject to authorization, MTU, latency, cost, and reliability constraints. This is the "water finding another opening" intuition: scale out before overflow, retain hysteresis to avoid route flapping, and keep a deterministic fallback if forecasts or telemetry are missing. It is a testable control hypothesis, not a guarantee that prediction will prevent every interruption.

The sink analogy is intentionally limited: unlike water outlets, network paths have distinct cost, delay, MTU, trust, and correlated-failure properties. The controller must choose eligible paths and reserve enough headroom rather than simply activate every available medium.

For each eligible path r, define a non-negative score s_r based on capacity, latency, forecast pressure, positive pressure derivatives, and operator-configured cost. One illustrative score is:

$$s_r = \frac{C_r}{D_r + \epsilon} \frac{\exp[-(\lambda \hat{p}_r + \mu \max(0, \hat{v}_r) + \nu \max(0, \hat{a}_r))]}{1 + \kappa K_r}.$$

Here C_r is a path capacity estimate, D_r a latency estimate, K_r normalized cost, and the coefficients are policy parameters that require calibration. Traffic shares are $\alpha_r = s_r / \sum_j s_j$ across the selected paths, subject to path availability, MTU, per-path capacity, and service constraints. This score is a starting point for comparison, not a globally optimal solution. A practical optimizer must account for shared bottlenecks, flow ordering, reordering buffers, deadlines, loss recovery, energy, and operator objectives.

The path set determines packetization. For each classical route, the usable payload MTU is the minimum link MTU minus encapsulation overhead. If the route set changes, packetization and reassembly rules must remain explicit; quantum frame capacity is not a classical byte MTU. A deployed design should use standardized transport and path-MTU mechanisms where possible, and test fragmentation behavior across tunnels and heterogeneous domains.

Session continuity is defined here as retaining application-level identity and recovering the intended

data despite path changes, buffering, retries, or reordering. It is not a guarantee of zero delay, zero packet loss, or uninterrupted physical connectivity.

6. Self-scaling content and AI placement

The CDN layer learns regional demand and estimates which content is likely to be requested. A cache or workload-placement action should be evaluated by its expected service benefit, not demand alone. An illustrative net benefit for replicating object o at node n is:

$$\begin{aligned} B_{o,n} = \mathbb{E}[&\text{miss cost avoided}] \\ &- \text{replication transfer cost} \\ &- \text{storage and energy cost} \\ &- \text{policy and freshness risk.} \end{aligned}$$

Replicate only when $B_{o,n}$ exceeds a policy threshold and the content is authorized, verifiable, fresh, and eligible for that location. Similar constraints apply to AI inference placement: estimate end-to-end network delay, compute queueing, energy, model availability, accelerator compatibility, and data-governance requirements. Training and bulk model distribution may be scheduled opportunistically; interactive inference may need a nearby node or a local fallback.

AI is not inherently safe, accurate, or economical. A model can produce a forecast, but admission control, resource limits, rate limits, and fail-safe routing should remain enforceable without model cooperation. Cache integrity requires cryptographic publisher authorization and trustworthy key management; a local checksum alone does not establish publisher identity.

7. Orbital AI data centers: a constrained edge tier

Orbital data centers are an intriguing but speculative extension. The near-term research question is not whether to move terrestrial hyperscale data centers into orbit, but whether a limited compute-and-storage payload can provide useful network-edge services for selected missions or regions.

Potential roles include:

- temporary processing of Earth-observation data before downlink;
- caching or inference for remote regions and space missions;
- relaying data across otherwise intermittent contacts;
- coordination among satellites and terrestrial edge/cloud facilities.

These are candidate use cases, not demonstrated benefits of the proposed architecture. Their value depends on workload demand, downlink scarcity, constellation geometry, alternative terrestrial capacity, and lifecycle cost.

Orbital compute faces constraints unlike terrestrial data centers. Power is limited and time-varying; electronics must tolerate radiation and thermal cycling; heat must be managed through spacecraft

thermal-control systems; maintenance and replacement are difficult; launch mass is costly; and connectivity to users or cloud regions is intermittent and geometry-dependent. NASA's small-spacecraft technology reference describes spacecraft thermal control as a balance of absorbed, stored, generated, and radiated heat [5]. Thus, abundant solar input does not imply unlimited usable compute: conversion, storage, eclipse operation, radiator area, pointing, and thermal limits all constrain the service envelope.

Optical communication can support high-rate space links under suitable pointing, acquisition, tracking, and weather conditions. NASA's Deep Space Optical Communications project is a technology demonstration, not evidence that a global orbital data-center network is already available [6]. System studies must include link availability, acquisition failures, ground-station weather, spectrum and licensing, and the effects of intermittent contacts.

Space sustainability is a first-order design constraint. Additional orbital compute hardware must be assessed for conjunction risk, debris mitigation, post-mission disposal, servicing, and long-term orbital capacity. UNOOSA documents international space-debris mitigation work and guidelines [7]. Energy and environmental comparisons must use a full lifecycle boundary: manufacture, launch, operation, replacement, downlink, and end-of-life—not orbital operating power alone.

Accordingly, the proposed scheduler treats orbital nodes as optional, failure-prone resources. It should select them only when their expected end-to-end service benefit exceeds terrestrial or aerial alternatives while satisfying safety, thermal, power, and policy constraints. Delay-tolerant applications may tolerate contact gaps; interactive applications may not.

8. Quantum networking boundary

Quantum networking may eventually contribute specialized services such as quantum key distribution or entanglement distribution, depending on protocols and infrastructure. It should not be modeled as an ordinary path carrying application bytes. Classical coordination, authentication, and user payloads still require classical channels. Qubit-frame capacity and key-resource availability are separate quantities from Ethernet/IP MTU and classical throughput.

The repository's quantum abstraction is only a simulated resource buffer; it does not implement QKD, entanglement routing, or cryptographic protection. Any future system must identify a specific quantum protocol, threat model, classical-control requirements, and security proof before making security claims.

9. Research and evaluation agenda

The vision should be evaluated incrementally against reproducible baselines:

1. **Trace-based forecasting:** compare the second-order predictor with persistence, moving-average, and appropriate learned baselines using temporally separated real link traces. Report

MAE/RMSE, calibration, prediction lead time, and degradation under missing or delayed telemetry.

2. **Multipath transport:** compare single-path, static striping, and congestion-aware active-active policies under replayed path delay, capacity, loss, and failure. Measure goodput, p50/p95/p99 completion time, reordering, retransmissions, continuity success, and energy/cost per delivered byte.
3. **Cache placement:** compare no-cache, reactive LRU, demand-only, and forecast-aware admission. Measure hit ratio, origin load, replication bytes, cache churn, freshness errors, and cost. Include high-cardinality and adversarial request workloads.
4. **Safety and robustness:** test stale or manipulated telemetry, corrupted content, model shift, NPU/runtime failure, control-plane partition, and unavailable links. Require bounded behavior and deterministic fallback.
5. **Orbital scenarios:** use documented orbital/contact schedules and link-budget assumptions. Compare orbital compute against terrestrial and aerial alternatives, including power, thermal limits, storage, downlink, launch and replacement assumptions, and end-of-life constraints.
6. **Testbed progression:** move from simulation to a controlled network emulation, then a small authorized multi-node testbed. Verify interoperability and operator controls before any public-network experiment.

Every evaluation should disclose topology, trace provenance, model hyperparameters, random seeds, software versions, baseline tuning, confidence intervals, and failure cases. Claims of lower cost or environmental impact require explicit and comparable accounting boundaries.

10. Synthetic policy comparison

To make the tradeoffs concrete, a reproducible synthetic benchmark compares three explicitly defined policies across four scenarios. These are **not** implementations of 5G or 6G standards, vendor products, or deployed networks. "5G-oriented single-path baseline" denotes a deliberately constrained single-path, fixed-cache reference; "reactive multipath baseline (6G-oriented)" denotes a reactive multipath reference; and "proposed predictive-cooperative policy" denotes the forecast-assisted policy described below. The graph/CSV labels identify experiment-specific policies, not implementations of mobile generations. The comparison cannot establish that one radio generation is better than another.

This qualification matters: simultaneous multipath is not absent from all current systems. MPTCP specifies concurrent paths for a connection [11], and 3GPP's 5G system architecture includes access-traffic steering, switching, and splitting mechanisms [12]. Their availability and behavior depend on device, transport, network, operator configuration, and the particular access combination; these standards do not imply that every device can bond arbitrary Wi-Fi, mobile, HF, and satellite paths into one mission-critical session. The comparison therefore tests explicitly defined control policies rather than asserting that all 5G devices are single-path. Likewise, the 6G-oriented profile is a research baseline, not a measured deployed 6G system; IMT-2030 work is a framework for future systems [2].

Each run simulates 180 steps of 30 seconds (90 minutes of synthetic time). Thirty independent demand-noise seeds are generated from seed 2026 and reused across policies for each scenario. The scenarios model an urban flash crowd over fiber, microwave, cellular, and LEO links; rural growth over microwave, cellular, and LEO links; mobility with link outages; and intermittent orbital contacts with microwave and GEO fallback. Capacities, latency, packet loss, availability schedules, link cost, and demand functions are synthetic parameters in `tools/benchmark_vision.py`. They are not fitted to operator data.

The 5G-oriented reference selects one currently available path using observed latency and keeps its initial cache-hit fraction. The 6G-oriented reference allocates traffic over eligible paths using current pressure, capacity, latency, and modeled cost, and adapts cache hit toward a recent demand average. The proposed predictive-cooperative policy uses the same allocator, forecasts normalized path pressure over three steps (90 seconds), forecasts demand without clipping ratios above baseline, and allows faster bounded cache warming. The current benchmark therefore does **not** simulate the full active-active topology reorganization, learned AI controller, failure-driven route/cache/compute recovery, or Kubernetes/multicloud orchestration described in the vision. Its predictor is a hand-coded heuristic; using "AI" for these particular graph results would overstate what was tested. The simple policies isolate predictive routing and cache adaptation only; they are not tuned production controllers.

Reported metrics are mean delivered goodput; delivered-traffic-weighted p95 latency, including a fixed 2 ms cache-service latency; the percentage of steps delivering at least 95% of requested demand; mean cache-hit fraction; and mean modeled share of demand served from cache (not an object- or request-level hit ratio). Modeled cost per delivered GB combines configured per-link transfer prices with $0.07 per newly replicated GB; it excludes storage, energy, cache eviction, and infrastructure costs. Bars show means and normal-approximation 95% confidence intervals (mean ± 1.96 standard errors across 30 seeds). The intervals describe variation between these stochastic runs, not uncertainty about real networks or a significance test of paired policy differences. The full machine-readable output, including intervals, is `figures/vision_benchmark_summary.csv`.

Scenario	Goodput (Mbps), single-path / reactive multipath / predictive-cooperative	Continuity (%), single-path / reactive multipath / predictive-cooperative
Urban flash crowd	176.4 / 298.1 / 297.5	10.5 / 100.0 / 100.0
Rural coverage and growth	38.8 / 73.8 / 74.0	0.0 / 100.0 / 100.0
Mobility and link outages	119.5 / 148.6 / 148.7	45.8 / 100.0 / 100.0
Intermittent orbital contacts	71.8 / 102.2 / 103.3	50.0 / 95.0 / 100.0

Across these parameter settings, predictive caching increased the mean cache hit fraction relative to the reactive reference in every scenario (for example, 25.4% versus 13.0% in the urban case).

Modeled cost per delivered GB was also lower than the reactive reference in all four cases: $0.0266 versus $0.0321 (urban), $0.0512 versus $0.0667 (rural), $0.0318 versus $0.0354 (mobility), and $0.0811 versus $0.1007 (orbital). Goodput was nearly unchanged between the predictive and reactive multipath policies in the first three scenarios; the largest relative increase was about 1% in the orbital scenario. The orbital-policy p95 latency was approximately 303 ms versus 477 ms for the reactive policy, while their continuity was 100.0% and 95.0%, respectively. These results show a modeled cache/cost and orbital-continuity tradeoff, not a general performance gain. In particular, single-path had lower urban p95 latency (26 ms versus 58 ms for both multipath policies) but much lower continuity. The predictive policy also cost more than the single-path reference in every scenario because it uses additional paths and cache replication.

The following side-by-side results make the intended distinction and the current model's limits explicit:

Synthetic comparison	Reactive 6G-oriented baseline	Proposed predictive/cooperative policy	Interpretation
Urban goodput	298.1 Mbps	297.5 Mbps	Essentially equal; both use the same path allocator
Urban mean cache-hit fraction	13.0%	25.4%	Predictive cache warming is the clear modeled difference
Urban modeled cost	$0.0321/GB	$0.0266/GB	Lower under the chosen synthetic transfer and replication prices
Orbital service continuity	95.0%	100.0%	Improvement in the scheduled-contact scenario only
Real-time video on-time, reactive vs predictive overflow	97.13%	97.72%	+0.59 percentage points under this workload
Real-time video on-time, predictive vs predictive + media assist	97.72%	99.18%	Additional gain from a separate duplication policy; 47.04% packet overhead and 1.862 Mbps duplicated payload

These are model outputs, not measured 6G versus 7G network performance. The last row includes a distinct media-assist policy and should not be attributed to predictive AI routing alone. In particular, the benchmark contains neither a trained AI controller nor failure-driven multicloud/Kubernetes recovery; those remain untested parts of the vision.

The plots below provide the complete KPI comparison and an illustrative single-seed urban trace. They are generated by running `py tools/benchmark_vision.py --output-dir figures --replicates 30 --seed 2026`.

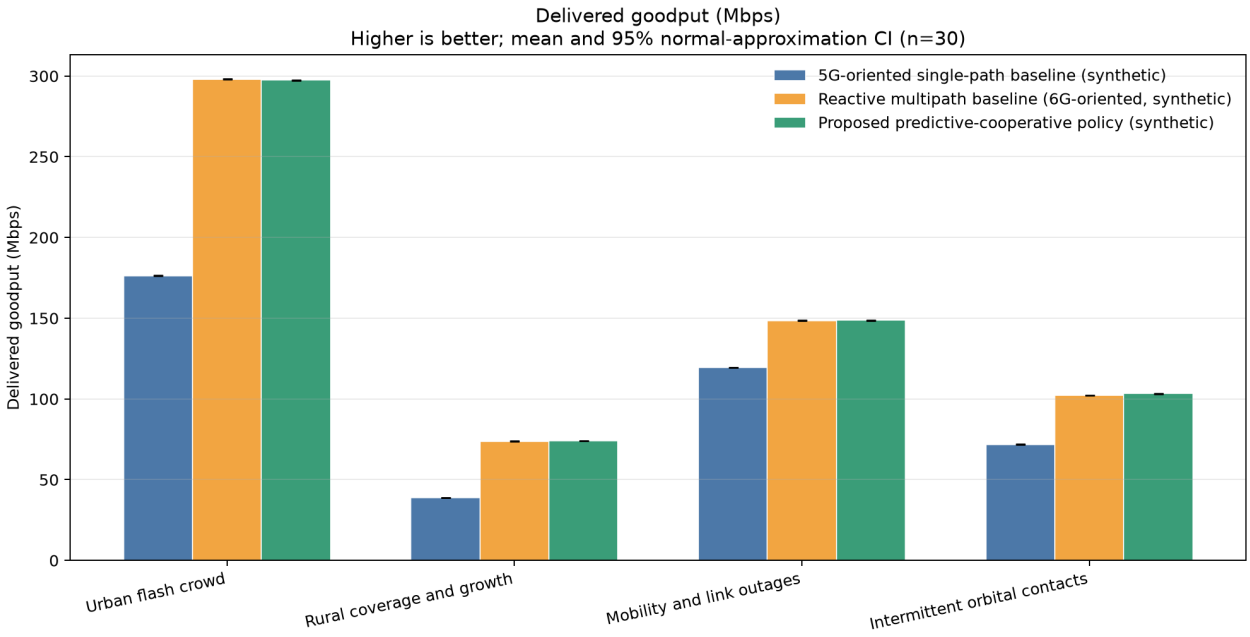


Figure 1: Delivered goodput by scenario and policy

These are synthetic policy experiments with hand-selected assumptions. In particular, the high single-path continuity failures are a consequence of the configured capacity constraints, not evidence about real 5G coverage; the reactive and predictive policies are abstractions rather than 6G/7G system implementations. Their near-identical goodput is expected because they share the same allocator and total path capacity. The tested proposal's primary differences are cache prediction, modeled cost, and selected orbital outcomes; it does not yet test the broader brain-inspired self-reorganization claim. The results are sensitive to cost weights, demand and outage schedules, cache limits, and policy parameters. They do not model transport retransmission, shared bottlenecks, packet reordering, radio scheduling, power, or physical infrastructure costs. Reproduction and extension should begin with the benchmark code and CSV, then replace synthetic assumptions with documented traces and validated policy baselines before making operational claims.

10.1 Public-safety handover and congestion-wave study

A second experiment focuses on the stated continuity problem: an application moving between Wi-Fi, mobile, HF, and satellite while traffic demand accelerates toward a path bottleneck. A

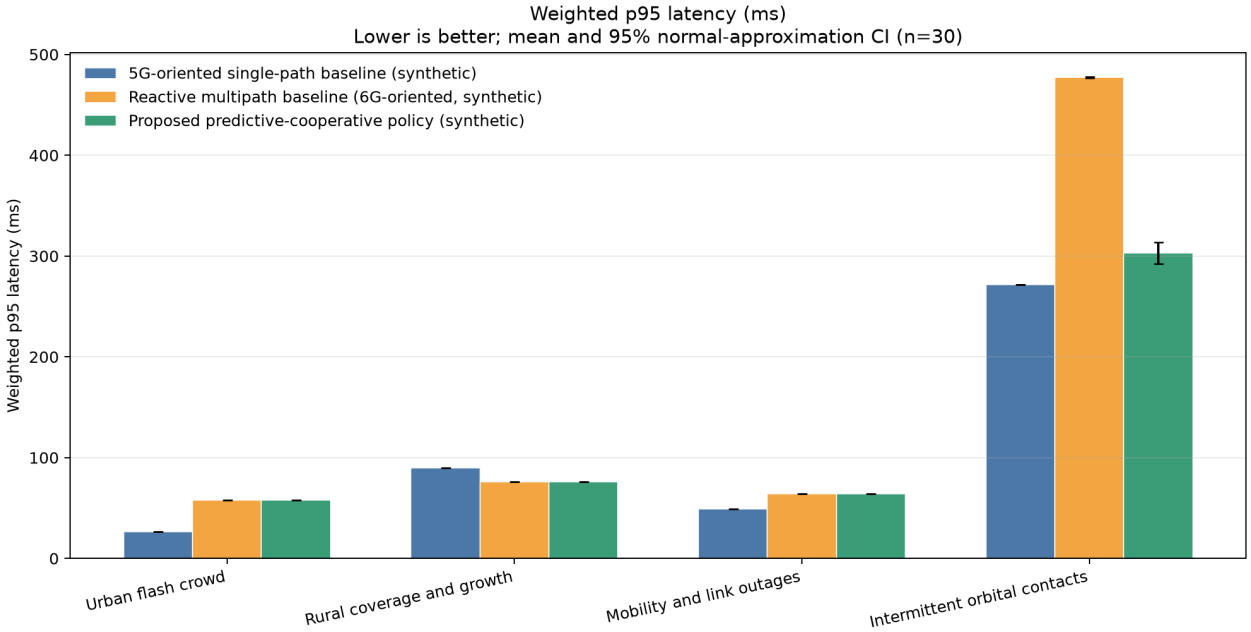


Figure 2: Weighted p95 latency by scenario and policy

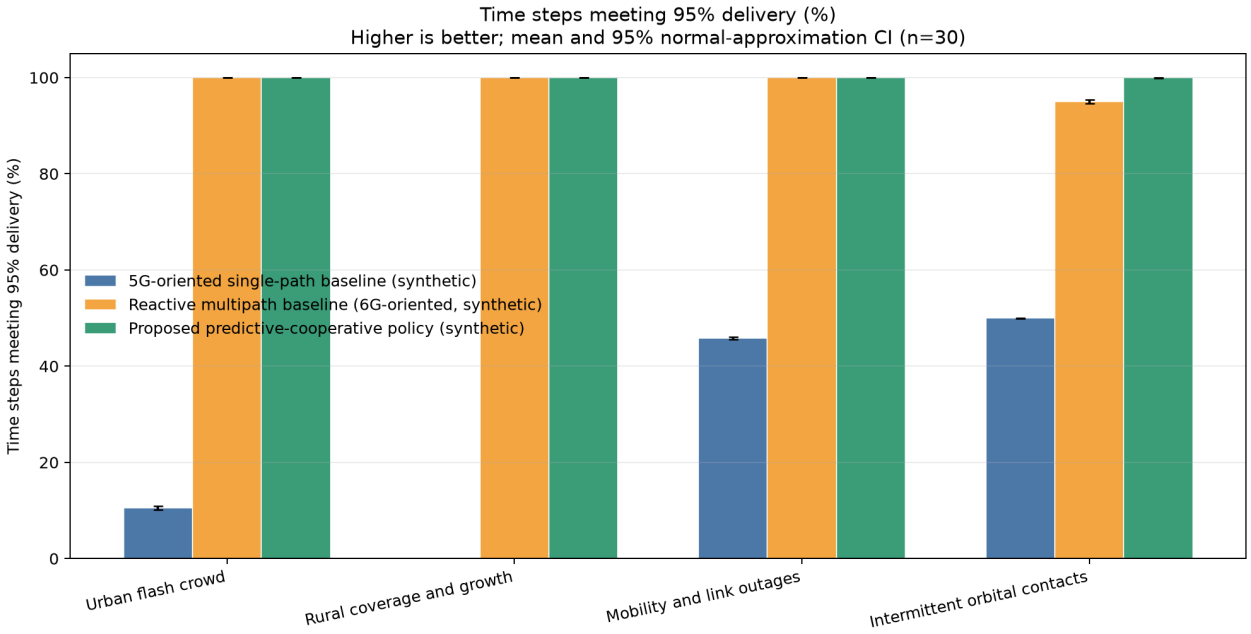


Figure 3: Service continuity by scenario and policy

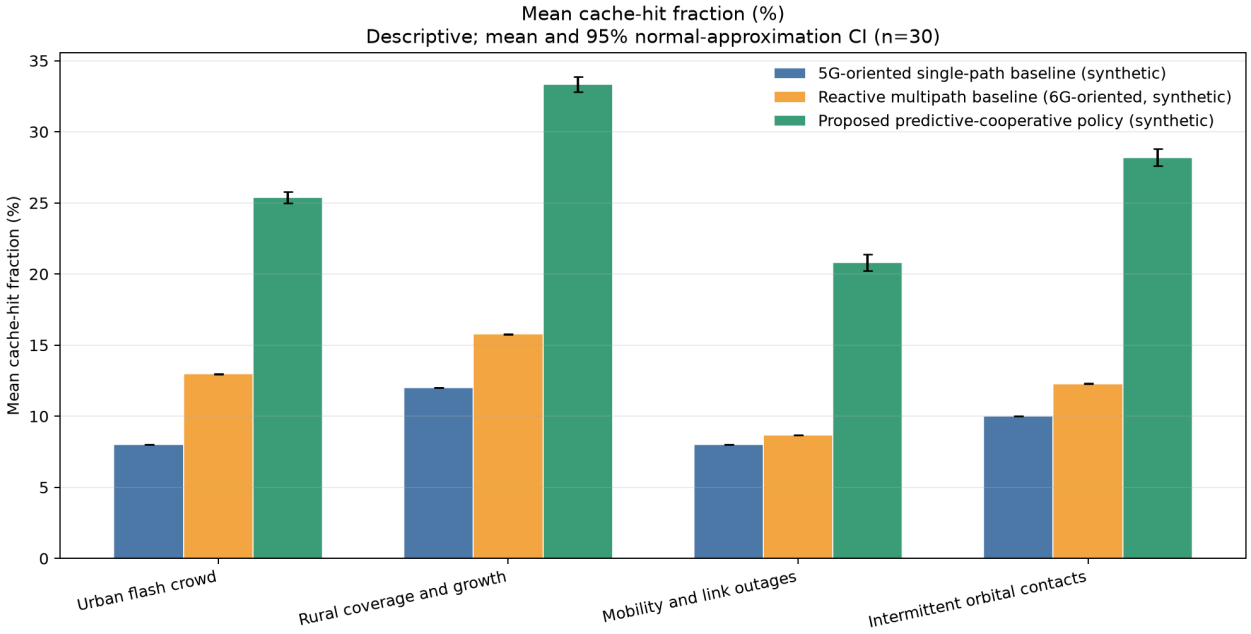


Figure 4: Cache-hit fraction by scenario and policy

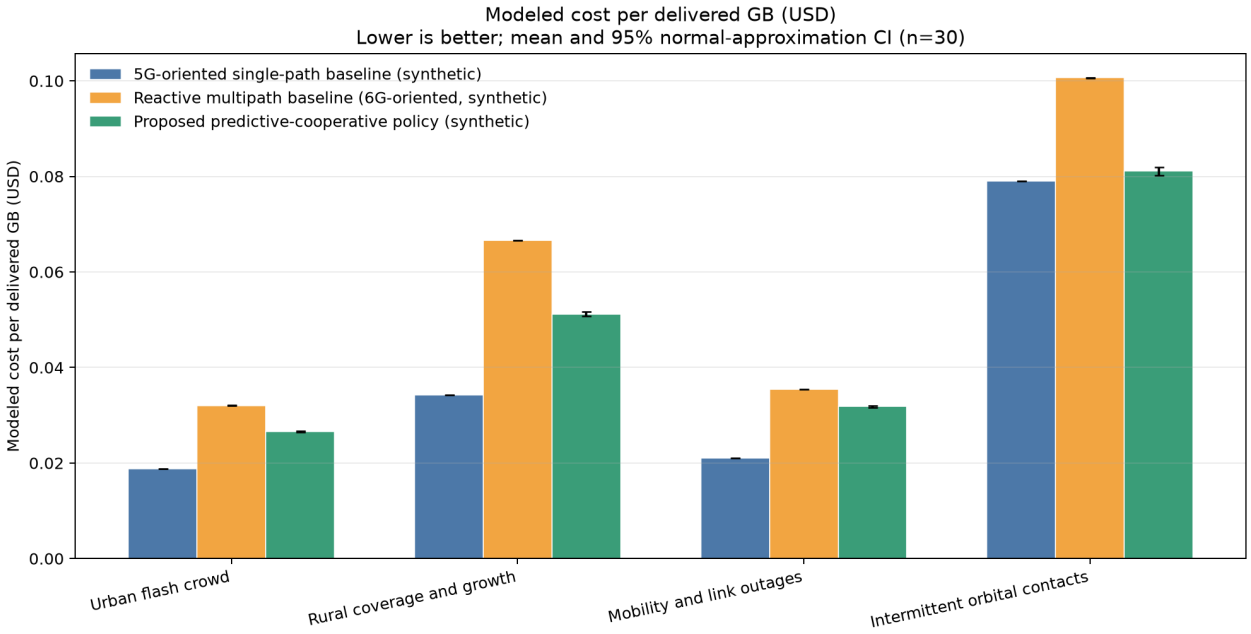


Figure 5: Modeled cost per delivered GB by scenario and policy

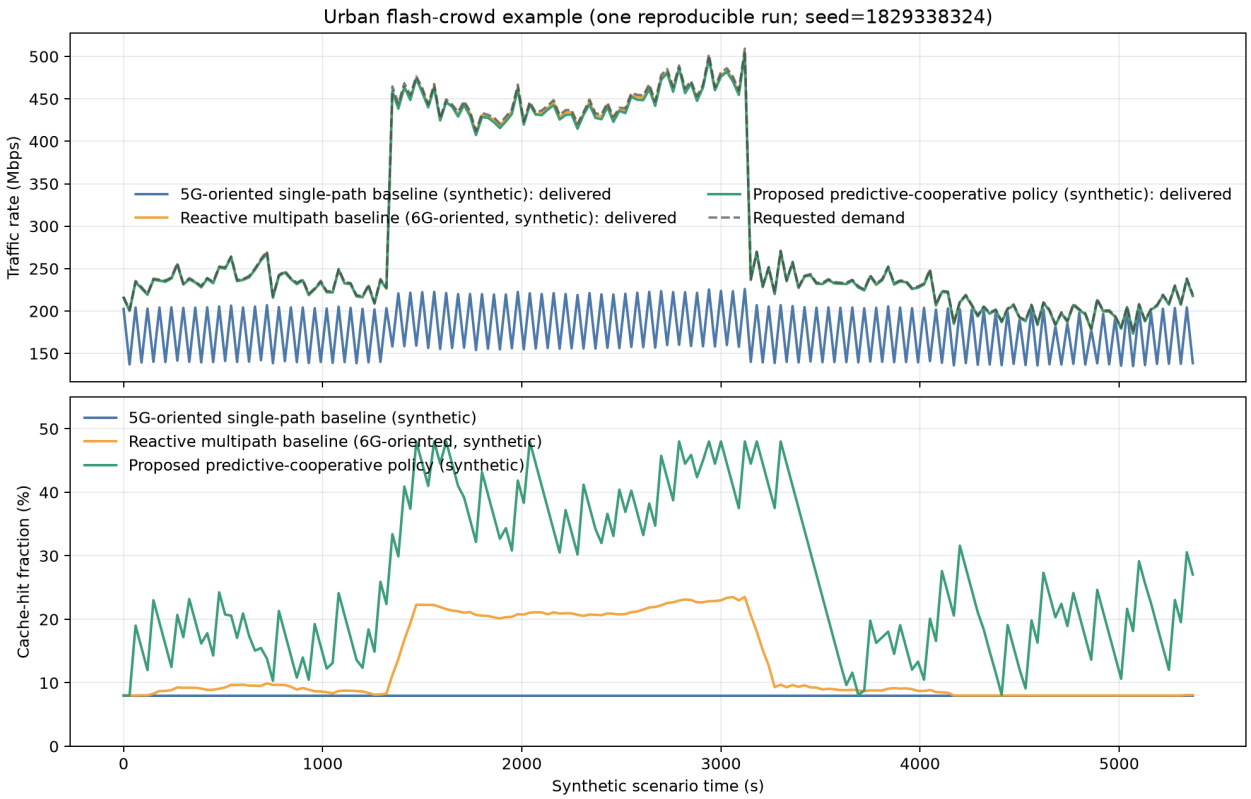


Figure 6: Example urban demand, delivered goodput, and cache-hit time series

180-second synthetic run uses one-second control steps, 30 random seeds derived from root seed 2026, 12 Mbps of prioritized mission-critical traffic, and a bursty bulk workload whose rate increases quadratically before two additional load waves. The initial path capacities are 80 Mbps (Wi-Fi), 150 Mbps (mobile), 60 Mbps (HF), and 280 Mbps (satellite). The Wi-Fi path is unavailable from second 50 through second 64. These are illustrative model inputs, not measured link or public-safety service characteristics.

The **single-path handover** baseline uses one medium and incurs a modeled two-second service gap when it changes paths at an 85% capacity threshold or after a path loss. The **reactive multipath** profile activates additional paths when current offered load reaches 90% of active aggregate capacity. The **predictive overflow multipath** profile uses a five-sample median of demand velocity and acceleration, bounds estimated acceleration to ± 0.1 Mbps/s^2 to limit noise amplification, and predicts six seconds ahead; it opens enough additional paths to keep forecast demand below 80% of active capacity. Active paths share traffic through the same bounded capacity allocator, with the critical class allocated first. This is a simplified policy comparison, not an implementation of MPTCP, 3GPP ATSSS, or a public-safety system.

The main outcomes are application-level multimedia QoS, not delivered goodput. In each one-second control step, the simulator produces 30 video frames per second, each represented by 14 independently transmitted segments, and 50 audio packets per second. A video frame is considered on time only if all segments arrive within a 150 ms deadline. Audio packets are duplicated only by

the predictive media-assist policy, over up to two active paths; other policies send one audio copy. A packet counts as on time if at least one eligible copy arrives within 50 ms. Path delay includes configured base latency, a utilization-dependent queue term, and random variation; packet loss is sampled independently. Video stall time is the sum of missed 1/30-second frame intervals, while a freeze event is a run of missed frames lasting at least 150 ms. Audio delay variation is the p95 absolute difference between consecutive received-copy delays. The p95 frame delay is computed over reconstructed frames, including late frames but excluding frames with lost segments. The separate handover-gap metric counts control steps where a single-path transition has no selected route; it isolates modeled route-switch interruption from congestion-induced late or lost media. These deadlines and packetization are scenario assumptions, not certified application or public-safety requirements.

Thirty-replicate mean results are: Here, V/J denotes p95 reconstructed-video-frame delay and p95 audio delay variation, respectively. Media-assist packet overhead is the number of duplicate video-segment and audio-packet transmissions divided by the number of original media packets; it is a packet-count ratio, not a byte or capacity percentage.

Policy	Video on time	Audio on time	Stall total / max (s)	p95 V/J (ms)	Handover gap
Single path	43.64%	39.79%	101.44 / 95.93	138.6 / 9.21	4.0 s
Reactive	97.13%	99.19%	5.17 / 1.00	137.1 / 2.24	1.0 s
Predictive	97.72%	99.75%	4.10 / 0.07	137.3 / 2.25	0.0 s
Predictive + media assist	99.18%	99.90%	1.47 / 0.05	48.9 / 2.30	0.0 s

The single-path policy has two modeled transitions, each with a two-second interval with no selected path: four seconds total and 120 missed video frames at 30 frames/s. Reactive multipath has a one-second standby-path activation gap (30 frames), while the predictive policy has none under this schedule. The single-path's much larger cumulative stall and 95.93-second longest freeze also reflect later capacity/queue overload in the scenario; they must not be attributed to handover alone. Multipath avoids most or all of these modeled switch gaps. Acceleration forecasting improves video on-time delivery by 0.59 percentage points and audio by 0.56 points versus reactive multipath, and reduces cumulative stall by 1.07 seconds. Its p95 video delay and audio variation are effectively unchanged, so the predictive policy alone does not show a broad QoS advantage over reactive multipath.

Adding media assist improves video on-time delivery by a further 1.46 percentage points and audio by 0.15 points versus predictive multipath, while reducing total stall by 2.63 seconds and p95 reconstructed-frame delay by 88.3 ms. The tradeoff is 47.04% duplicate packets relative to original media packets and a mean 1.862 Mbps of duplicated media payload in this scenario. The configured source media rates are 4 Mbps video and 0.128 Mbps audio; the duplication rate measures selected

packet copies, not the full source rate. Duplicate payload load contributes to modeled utilization for subsequent media packets, but the simplified allocator does not feed it back into bulk-flow allocation. These are synthetic QoS and resource-model results, not evidence that a deployed transport accelerator would achieve the same gains.

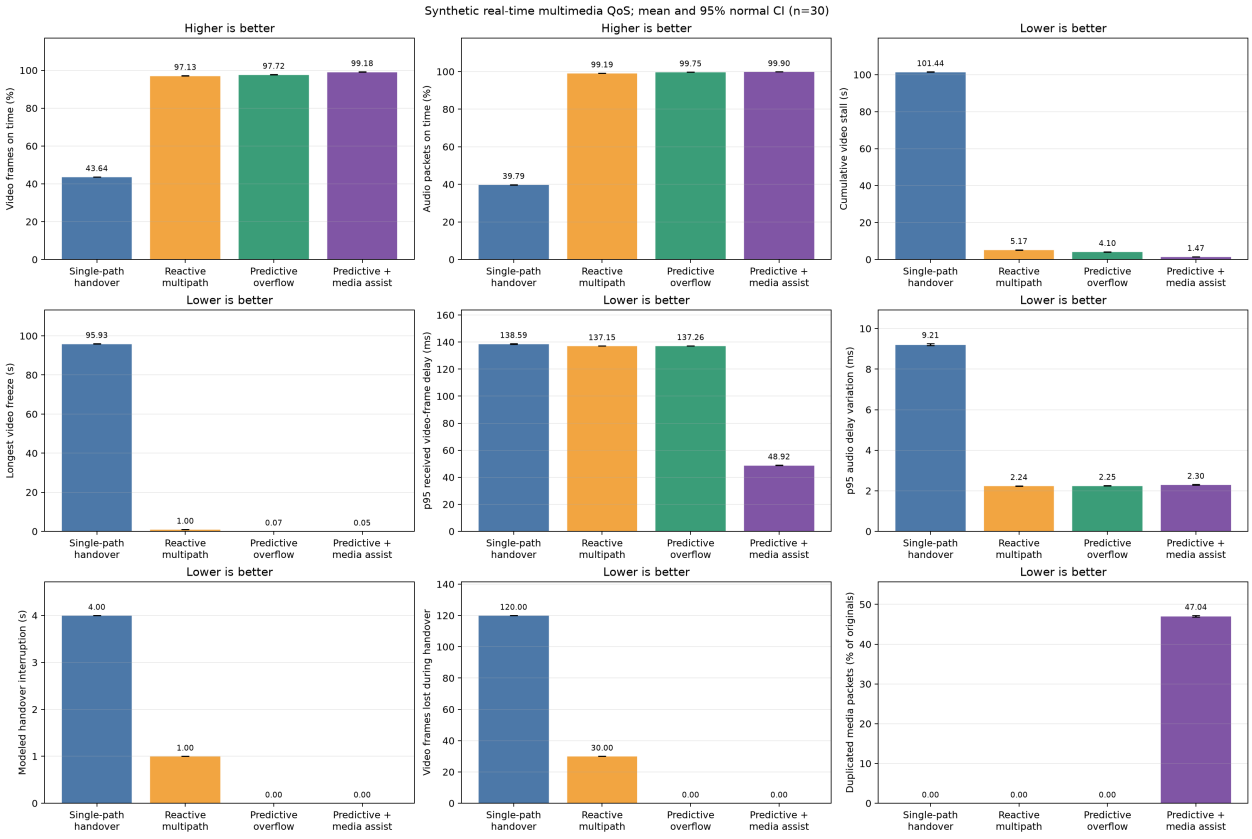


Figure 7: Real-time multimedia QoS: frame and packet delivery, stalls, delay, jitter, and handover interruption

The figure and CSV report normal-approximation 95% confidence intervals across the 30 seeds. These results illustrate the mechanism under the stated assumptions; they are not deployment measurements or an estimate of real public-safety availability. In particular, multipath activation is modeled without path setup delay, the synthetic transport omits protocol retransmission and reordering, and the Wi-Fi, mobile, HF, and satellite path characteristics are not measured. A real evaluation must measure app-level voice/video continuity, one-way latency, jitter, deadline misses, recovery time, and correlated failure using a fair 5G multi-access baseline where available, plus operator-approved 3GPP mission-critical service requirements [13]. The CSV with replicate-level summary statistics is `figures/public_safety_benchmark_summary.csv`.

10.2 Cooperative path and multicloud reorganization experiment

To test more of the proposed system than the earlier routing/cache benchmark, a separate Python experiment compares a **6G-oriented reactive multipath baseline** with a **proposed cooperative**

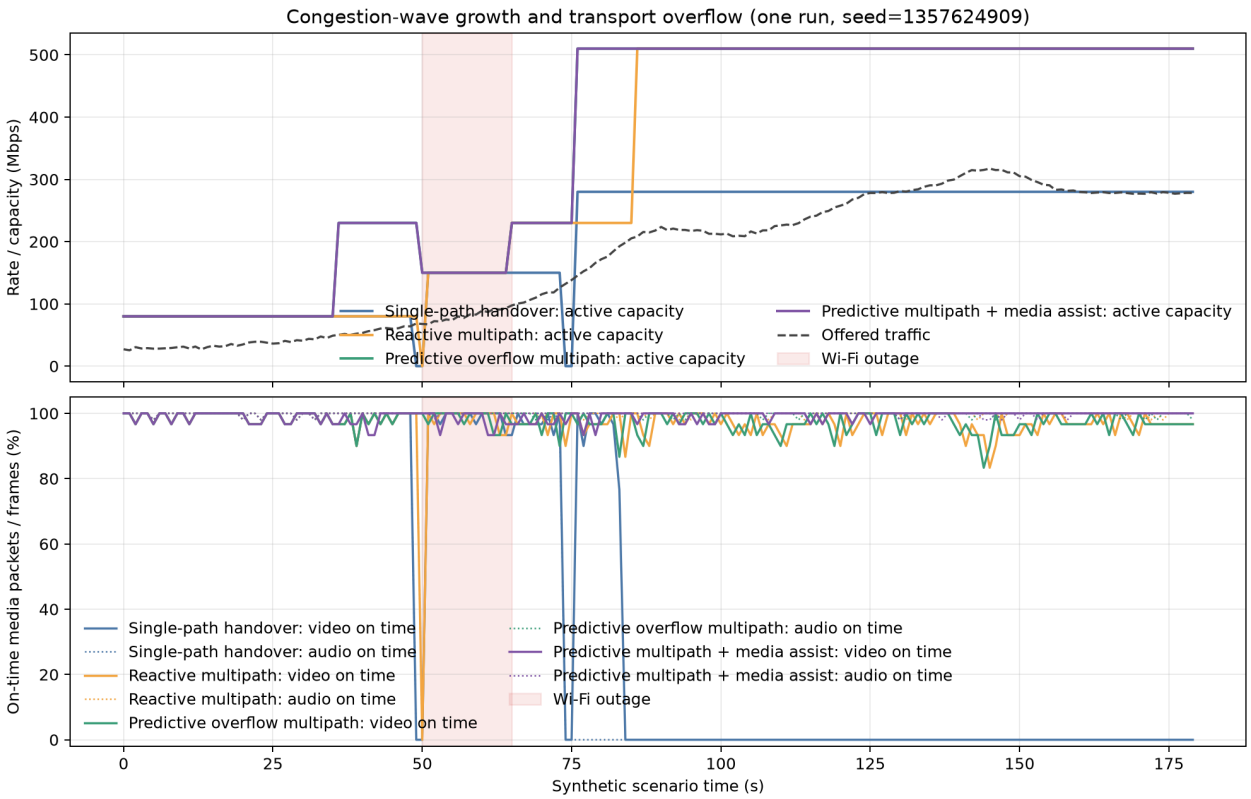


Figure 8: Congestion-wave capacity and on-time real-time media delivery over time

recovery policy. Both use the same heterogeneous paths, workload trace seeds, traffic demand, and failure schedule. The scenario lasts 240 seconds and includes a degrading/failing fiber path (degradation begins at second 70; outage from seconds 95–114), a mobile-path outage later in the run, an accelerating congestion wave, and a cloud-region failure. The path controller activates standby routes at a reactive 82% threshold with a three-second modeled setup delay. The cooperative policy uses an eight-second linear demand forecast, a 72% activation margin, and a one-second modeled setup delay; local path health deterioration can also trigger pre-activation. In this experiment, reactive multipath sends one copy of each media packet; selective duplication is enabled only for the proposed cooperative policy when its local risk conditions are met.

The cloud portion represents two Kubernetes-managed regions at an abstract capacity level, not by running Kubernetes. Region A degrades before becoming unavailable from seconds 155–179. The reactive baseline begins a cold standby-region start only after the failure and models a 20-second delay. The cooperative policy starts a 240 Mbps secondary region when region-A health begins declining; its assumed eight-second startup completes before the outage. Its secondary-region capacity-time is counted so this is not treated as free. Both policies use the same path and cloud capacities; they differ in forecast horizon, trigger margin, route setup, and replica preparation. Thirty paired demand seeds are used (root seed 2026).

Metric (synthetic mean, 30 replicates)	Reactive multipath baseline	Proposed cooperative policy
Demand delivered	89.21%	98.79%
Critical-service availability	91.67%	100.00%
Video frames on time	91.47%	99.80%
Audio packets on time	91.47%	99.80%
p95 received video delay	9.88 ms	9.64 ms
Service recovery after primary-path disruption	0.00 s	0.00 s
Recovery after cloud-region failure	20.00 s	0.00 s
Mean active paths	3.28	3.35
Duplicate media packets	0.00%	1.90%
Secondary-region capacity-time	1,200 Mbps · s	7,440 Mbps · s

Under these exact assumptions, the proposed policy improves modeled application delivery chiefly because the prepared cloud replica avoids the baseline's 20-second region-recovery interval. Both policies already meet the 95% delivery threshold at the instant the primary path fails, so the corrected path-service recovery interval is zero for both; there is no path-recovery advantage in this run. The cooperative policy uses more secondary-region capacity and a small amount of media duplication. The reported capacity-time is a resource exposure, not a monetary cost. These outcomes depend on the assumed warning lead time, startup delays, controller thresholds, and resource budget. A 6G deployment could also implement pre-warmed multicloud replicas or faster setup; the comparison tests these policies, not a limitation of 6G as a technology.

The zero-duplication baseline is an explicit experiment setting, **not a statement that 6G networks cannot duplicate packets or provide multipath redundancy**. It makes the chart a comparison of these two configured policies; the proposed policy's 1.90% is a modeled overhead cost, not by itself evidence of an advantage. A stronger media-specific study should add a reactive baseline with the same duplication budget and compare selective prediction against that baseline on identical traces.

The paired-seed differences were also recomputed directly from the 30 seeded runs. Cooperative minus reactive demand delivery was +9.58 percentage points (normal 95% paired interval +9.57 to +9.59), critical-service availability was +8.33 points, and video on-time delivery was +8.33 points (interval +8.32 to +8.34). Cloud recovery was 20 seconds shorter under the configured startup assumptions; service recovery at the fiber failure was zero for both. These narrow intervals quantify only the small randomized demand variation around one fixed synthetic scenario. They do not cover uncertainty in capacities, failure timing, warning lead, startup time, model error, or baseline tuning, and therefore are not evidence of real-network effect size.

This benchmark is a controlled software experiment, not an empirically validated network model. Its policies change several factors at once: forecasting, activation threshold, setup delay, replica preparation, and media duplication. The cloud benefit is directly conditional on an eight-second prepared-replica startup before the modeled outage versus a twenty-second reactive cold start. In this experiment, each video frame is represented by one sampled packet rather than a segmented encoded frame; duplicate traffic can affect modeled media utilization/delay but is not fed back into bulk allocation or the system-capacity delivery metric. Consequently, the results support only the statement that this particular cooperative rule set performs better under its specified assumptions. They do not isolate an AI effect, prove a brain-inspired mechanism, establish a 6G limitation, or show superiority over a real 6G deployment.

The experiment is intentionally a **functional abstraction**, not a full distributed-AI or Kubernetes simulation. The "agents" are represented by local link/cloud health observations coordinated by deterministic forecast-and-threshold logic; no trained neural controller, reinforcement learning, inter-agent messaging, NPU execution, live Kubernetes API, or measured 6G trace is used. It models failover, not savant-like cognitive mechanisms. The 95% confidence intervals and all per-policy values are in the benchmark CSV; the comparison chart is shown below.

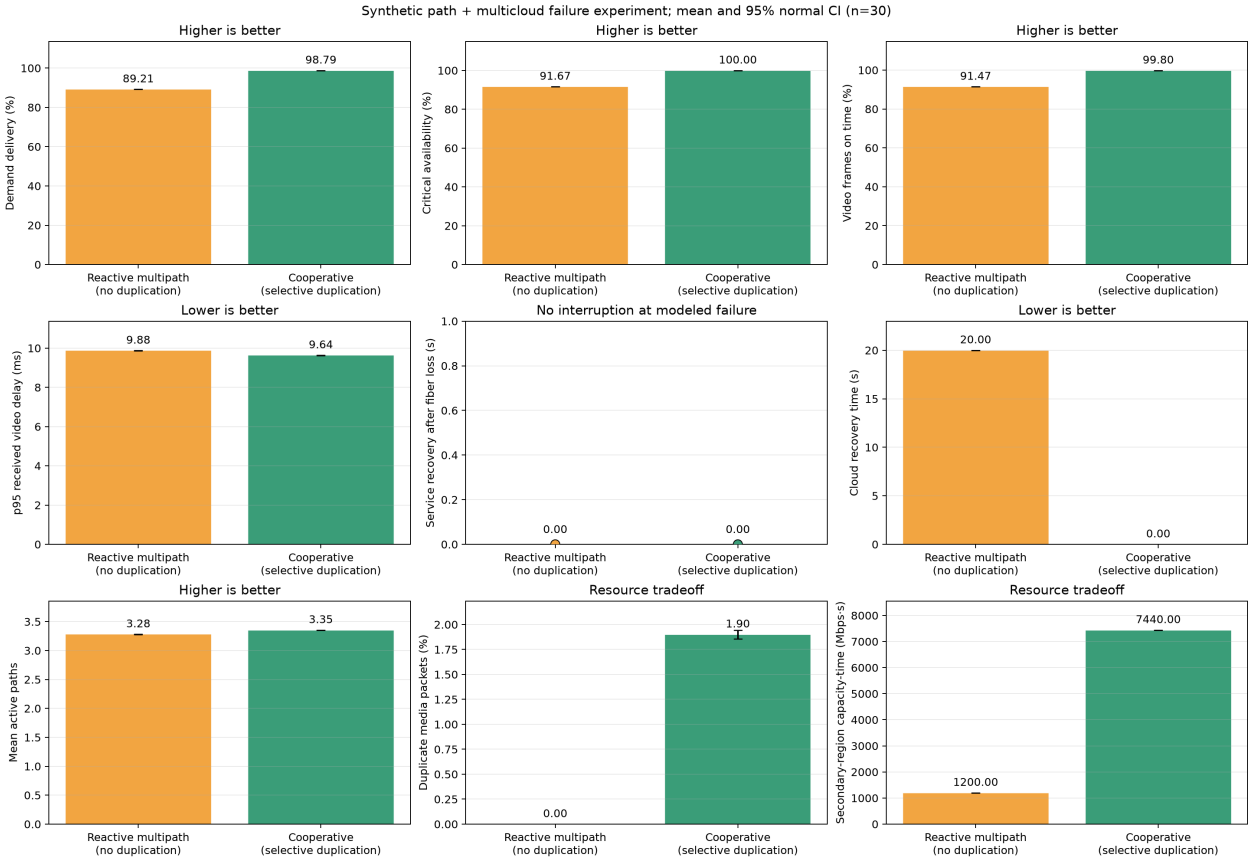


Figure 9: Synthetic reactive multipath versus cooperative path/cloud recovery

The default script settings reproduce the 30-replicate run with root seed 2026; the README gives the full command.

11. From multipath handover to multicloud Kubernetes

The proposed system spans two different adaptation timescales. A real-time session controller must preserve media delivery while its usable transport changes; a service-orchestration controller must place and scale caches, relays, and application workloads across edge sites and cloud regions. These controllers exchange goals and capacity signals, but must not be collapsed into one loop: a Kubernetes pod start is far too slow to repair a packet-level handover, and rapid path steering must not create unbounded cloud replicas.

11.1 Session and transport continuity

An application session has a stable service identity independent of its current access bearer. A session endpoint or trusted edge gateway measures per-path delay, jitter, loss, available capacity, MTU, reachability, and confidence in the telemetry. A path manager maintains active and standby subflows across eligible media such as Wi-Fi, cellular, private radio, HF, and satellite. Authentication, application policy, and path MTU are checked before a path is admitted. Multipath transport such as MPTCP can use several IP paths for one byte stream where both endpoints and the network support it [11]; 3GPP access-traffic steering, switching, and splitting is a relevant 5G-system capability [12]. Neither capability implies universal support for arbitrary access combinations or identical handover behavior across devices.

The predictive controller estimates demand and path-pressure velocity and acceleration. If projected load approaches a safety margin, it can establish or reserve a secondary path and shift eligible traffic before the serving path saturates. During a failure, a pre-established path may avoid the setup gap modeled for single-path switching. Traffic may be striped when reordering and application semantics permit; deadline-sensitive audio may be duplicated over independent paths; video may use selective packet duplication or application FEC when a path forecast indicates elevated deadline risk. The policy must bound redundancy, account for its capacity cost, respect data residency and priority, and retain deterministic fallback if telemetry or inference fails.

11.2 What “TCP amplifier” should mean

“TCP amplifier” is not a standardized protocol term and should not be read as creating bandwidth. TCP retransmission can add head-of-line delay: a missing byte may hold later bytes from the receiving application. That behavior can be undesirable for interactive media even when TCP eventually delivers every byte. Much real-time audio/video uses deadline-aware transports over UDP, including RTP/WebRTC profiles, while QUIC provides both reliable streams and datagram extensions depending on the application. The correct mechanism depends on the media stack, not a generic TCP switch.

This paper therefore uses **transport accelerator** or **media assist** for a policy layer, not a device that amplifies TCP. Its functions may include:

1. classifying flows by authenticated service policy and deadlines;
2. pacing and prioritizing packets without starving control or other users;

3. steering or striping packets over established subflows;
4. selectively duplicating only packets whose predicted path risk justifies the extra bytes, or adding bounded application-layer FEC;
5. measuring late/lost frames, jitter, freezes, recovery time, and redundancy overhead, then reducing assist when path conditions improve.

For legacy TCP, a deployment could use endpoint-supported MPTCP or an explicitly configured transport proxy. A split-TCP proxy changes end-to-end transport semantics and may conflict with end-to-end encryption, congestion control, authentication, and application assumptions; it must not silently acknowledge or discard data on behalf of an endpoint. This simulator models deadline-aware media packet duplication and path selection only. It does not implement TCP, MPTCP, QUIC, WebRTC, FEC, proxying, or real packet forwarding.

In the public-safety synthetic benchmark, predictive multipath with media assist is compared with predictive multipath alone. At root seed 2026 over 30 replicates, video frames on time increase from 97.72% to 99.18%, audio packets on time from 99.75% to 99.90%, and total video stall falls from 4.10 s to 1.47 s. p95 received video-frame delay falls from 137.3 ms to 48.9 ms, while p95 audio delay variation changes from 2.25 ms to 2.30 ms. The modeled assist duplicates 47.04% as many media packets as original packets and averages 1.862 Mbps of duplicated payload in this scenario. This is a QoS/overhead tradeoff under synthetic assumptions, not a claim that a real transport accelerator achieves these values.

12. AI- and NPU-assisted TCP transport acceleration

"TCP amplification" is best treated as a proposed service name, not a literal increase in a TCP connection's capacity. A neural processing unit cannot add link bandwidth, remove propagation delay, or make lost bytes arrive sooner. The technically meaningful goal is **AI-assisted transport optimization**: use measured and predicted conditions to select a safe combination of path, pacing, scheduling, and application-level protection while preserving TCP's reliability and congestion response. TCP behavior is defined by the standardized transport protocol, and congestion control is an essential response to shared bottlenecks [16, 17]. An accelerator must not defeat those controls by injecting acknowledgements, hiding loss, or encouraging a sender to exceed a safe rate.

Functional architecture

As introduced in Section 11.2, the design separates three functions that are often conflated by the word "NPU":

1. **Packet-forwarding hardware**—a switch/router ASIC or network processing unit (NPU) classifies, queues, schedules, filters, and forwards packets at line rate according to installed policy.
2. **Neural inference hardware**—a neural NPU, GPU, or CPU evaluates a compact model over summarized telemetry. It recommends path weights, a standby activation, a pacing envelope, or selective media protection.

3. **Endpoint transport**—the TCP sender and receiver retain sequence, acknowledgement, retransmission, and congestion-control state. A network controller may configure supported endpoint or multipath behavior, but it does not silently replace endpoint semantics.

The neural model should generally operate on a control interval (for example, hundreds of milliseconds to seconds), not infer separately for every packet. Packet actions need predictable latency and remain in the router's deterministic forwarding pipeline. A model can consume windowed features such as RTT and RTT variation, ECN marks, queue occupancy and trend, loss and retransmission rates, ACK spacing, path availability, offered load, and application class. It emits bounded recommendations such as "prepare path B," "reduce this flow's pacing ceiling," or "duplicate only these deadline-bound datagrams." The policy engine checks authorization, topology, congestion, capacity, and service constraints before programming the forwarding hardware or a supported endpoint API.

Where AI may help—and where it must not

Prediction may help distinguish a rising queue from a transient sample, anticipate a path outage, rank eligible paths by deadline risk and cost, or recommend a cautious pacing profile. For MPTCP-capable endpoints, supported subflows can be scheduled according to policy; RFC 8684 specifies the endpoint-level multipath TCP extensions [11]. For a conventional single-path TCP flow, changing the route may alter RTT, reordering, and failure behavior. An in-network device should not assume it can transparently split or migrate that connection while retaining semantics. An explicit proxy creates separate transport legs and therefore changes the end-to-end trust and failure model.

The controller must not treat every loss as proof of congestion or every idle queue as permission to increase the sending rate. Wireless fading, mobility, buffer overflow, route changes, and correlated failures can produce different signals; an incorrectly aggressive model can worsen congestion for other flows. Consequently:

- congestion-window evolution remains governed by a standards-compliant endpoint congestion controller (for example, CUBIC where implemented) [18];
- network pacing and prioritization respect aggregate link capacity, fairness, and configured service budgets;
- retransmission remains the responsibility of TCP endpoints; the network may provide telemetry but does not spoof ACKs or discard acknowledged data;
- packet duplication or FEC is applied only to application data whose deadline semantics support it, and its bytes are charged to the same capacity budget;
- policy is fail-safe: stale telemetry, model failure, overload, or accelerator unavailability returns control to the established non-AI forwarding and congestion-control path.

For encrypted traffic, classification must rely on authorized metadata, explicit endpoint signaling, or service policy rather than payload inspection. Model features and logs should avoid storing unnecessary user data. Signed/versioned models, provenance checks, sandboxed inference, rollback, operator override, and rate-limited actions are required before considering an autonomous deployment.

Training, placement, and Kubernetes integration

Training should use diverse measured traces spanning wired, cellular, Wi-Fi, microwave, and satellite conditions, including congestion, random loss, mobility, route changes, and correlated failures. Synthetic pretraining can initialize a model but cannot establish safe thresholds or generalization. Validation should use held-out networks, calibrated uncertainty, offline replay, and shadow mode before any recommendation is allowed to change live traffic. Online reinforcement learning must not explore by violating congestion, availability, or safety constraints; constrained contextual selection or offline policy evaluation is a safer initial research path.

In the multicloud architecture, regional Kubernetes clusters can deploy versioned model services and telemetry aggregators near routers. A global planner can distribute signed model artifacts and compatible policy bundles according to region, hardware capability, and data-residency rules. Local inference must continue when cloud connectivity is lost, and model rollout or pod autoscaling must not sit in the packet recovery loop. Kubernetes schedules the control-plane services and manages their lifecycle; router forwarding hardware performs the time-critical packet actions. Hardware accelerators must be explicitly inventoried and exposed through validated vendor drivers or device plugins—an NPU cannot be assumed merely because a cluster is Kubernetes-managed.

Evaluation criteria

An evaluation should compare a standards-compliant TCP congestion-control baseline, supported MPTCP scheduling where both endpoints permit it, static operator policy, reactive telemetry-based steering, and the proposed AI-assisted recommendation with deterministic safety guards. Replay paired traces with identical offered load and failure timing. Measure completion time, goodput, RTT distribution, retransmissions, ECN, queue delay, loss, fairness against competing flows, handover recovery, energy, inference latency, and model/action overhead. Include counterexamples: bad forecasts, model outage, route asymmetry, non-congestion loss, sudden bottleneck collapse, encrypted flows, and adversarial telemetry. Report whether QoS improves without harming other flows or violating congestion-control rules.

The current repository does not implement this TCP/NPU controller. It has a synthetic ONNX congestion forecaster and an optional inference-provider path, but no connection to TCP state, live switch queues, a forwarding ASIC, or Kubernetes model deployment. Accordingly, the concept is a research chapter and evaluation plan, not a demonstrated TCP amplifier.

13. Multicloud and Kubernetes control plane

Each edge, regional, and cloud location operates an independently managed Kubernetes cluster. A global placement service may observe summarized capacity, cache hit/miss demand, accelerator availability, service latency, egress price, trust domain, and residency constraints through authenticated cluster agents. It selects a target region or replica set; a local cluster controller then reconciles desired state into Deployments, StatefulSets, Services, storage claims, and node-local cache workloads. Kubernetes HPA or a custom-metrics scaler can adjust replica counts from queue

depth, request rate, or forecast demand, within configured min/max replicas and resource quotas. Topology-aware placement and affinity keep cooperating data-plane components close when that improves service.

The global service is not assumed to be a magical Kubernetes federation. Clusters have separate API servers, identities, failure domains, and networking. Cross-cluster service discovery, workload identity, policy, secrets, image/model provenance, service connectivity, and failover must be explicitly selected and validated. Cache placement is not the same operation as relocating a stateful service: cache objects can often be copied asynchronously after integrity and policy checks, while stateful workloads require consistency, ownership, and recovery semantics. The global planner should move content and pre-warm capacity ahead of demand where economical, but keep local serving and failover operational during cloud control-plane partition.

14. Coupling, safety, and evaluation

The end-to-end control loop is: (1) observe path and service QoS; (2) predict load and deadline risk; (3) admit an alternate authenticated path; (4) steer or selectively protect media; (5) report regional demand and resource pressure; (6) request cache replication or bounded Kubernetes scale-out; and (7) verify the resulting application QoS and cost. Path steering operates on packet-to-second timescales; orchestration and cache placement generally operate more slowly. Hysteresis, rate limits, budgets, and ownership boundaries are needed to prevent competing controllers from oscillating or scaling without limit. NPU inference may recommend actions, but authorization, rate limits, placement constraints, and fast failover remain deterministic and operator-controlled.

Evaluation should compare single-path handover, standards-capable multipath where available, reactive multipath, predictive multipath, and predictive multipath plus media assist. Replay shared Wi-Fi/mobile/HF/satellite traces with controlled outages and correlated failures. Measure session interruption duration, video frame deadlines, freezes, audio packet deadlines, jitter, tail latency, reordering, retransmissions or redundant bytes, energy, cost, and Kubernetes scale/placement delay. Include cases where duplication harms congestion, policies prohibit a route, or a cloud region is unavailable. The decision criterion is sustained application QoS within a resource and policy budget—not peak goodput in isolation.

15. Prototype status and limitations

The accompanying repository provides starting artifacts, not experimental evidence for planetary-scale operation:

Capability	Current prototype	Research gap
Multi-path mesh	In-process allocation plan using configured link attributes	Real forwarding, shared bottlenecks, transport behavior, and network testbed

Capability	Current prototype	Research gap
UDP packet path	Actual socket datagrams over local IPv4 loopback	Multi-host/NIC transmission, physical media, loss/reorder emulation, and scale
Congestion predictor	Small MLP pretrained on deterministic synthetic data	Training and calibration on measured, representative traces
NPU execution	Optional ONNX provider path with profile checks	Vendor hardware/runtime validation and physical-device evidence
CDN adaptation	Simulated demand forecasting, cache, and replica decisions	Production cache integration, freshness, privacy, and economic evaluation
Cooperative recovery	Deterministic paired path/cloud failure benchmark	Trained distributed AI, calibrated failure behavior, and real Kubernetes/cloud integration
Orbital infrastructure	Architectural concept only	Orbital dynamics, spacecraft subsystem model, mission cost, and field data
Quantum resource	Illustrative key-buffer abstraction	Protocol-accurate quantum network and security analysis

The current model's synthetic validation result is not a measurement of real-network accuracy. Likewise, a local UDP benchmark is not a measurement of wireless, satellite, microwave, or wide-area performance. The system has not been shown to improve global coverage, affordability, latency, resilience, or environmental impact.

16. Conclusion

Cooperative multi-medium backhaul offers a way to connect content placement, traffic engineering, and service continuity across terrestrial, aerial, and potentially orbital resources. The proposed temporal topology model gives the controller a shared representation of cross-layer services and constraints while leaving packet encapsulation and protocol interoperability to established standards and explicit adapters. The central proposal is a bounded, active-active control loop that uses telemetry trends to shift traffic and cache capacity before congestion becomes severe, while optimizing against explicit service, cost, energy, and policy constraints. Orbital AI edge nodes may be valuable for specific workloads, but their power, thermal, contact, cost, and sustainability limits must be treated as architecture inputs rather than future details.

This paper frames the idea as a research program, not a claim of a finished "7G" system. The path forward is measurement: validate the predictor, test multipath behavior under realistic impairments,

quantify content-placement tradeoffs, establish safe control boundaries, and evaluate orbital scenarios against terrestrial alternatives. Progress on those questions—not terminology alone—will determine whether this vision can contribute to more resilient and widely accessible communication.

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